

INFECTIVE ENDOCARDITIS — Clinical Revision Table (Dr Ali Jabari 2025)

Bold = high-yield. IE = microbial infection of heart valve (native/prosthetic), chamber lining, vessel, or congenital anomaly.

1 — DEFINITION, EPIDEMIOLOGY & PATHOPHYSIOLOGY

Epidemiology	5–15/100,000/yr; >50% over 60 yrs . Underlying: RHD 24%, CHD 19%, other (calcified AV, MVP) 25%, none 32%. Case fatality ~20% even treated (worse in PVE / resistant organisms)
Pathophysiology	Occurs at endothelial damage (high-pressure jets: VSD, MR, AR) → platelet/fibrin deposits colonized by organisms → vegetations (organisms+fibrin+platelets) → obstruction/embolism, abscess, cusp perforation, chordae rupture. Virulent S. aureus → normal heart ; TV endocarditis common in IDU. Low-pressure (large ASD) = minimal risk. Extracardiac: vasculitis, mycotic aneurysm, splenic/renal infarct, immune GN

2 — MICROBIOLOGY

S. aureus (incl MRSA) 31%	Acute, aggressive valve destruction; 50% of IDU cases ; mortality 25–40% (left heart)
Strep viridans 19%	Oropharyngeal, α-haemolytic, subacute , penicillin-sensitive
CoNS (epidermidis/lugdunensis) 10%	Prosthetic / foreign body, nosocomial
Enterococci 10%	GI flora, UTI/nosocomial; needs bactericidal synergy (+ gentamicin)
Strep bovis/gallolyticus 7%	GI flora — associated with colon polyps/cancer → colonoscopy ; subacute, PCN-sensitive
HACEK 2% / Culture-neg 14%	HACEK = fastidious G–ve, +culture >5d, large vegetations. Culture-neg : prior antibiotics, Coxiella (Q fever) , Bartonella (cat scratch) , Candida/fungi , Brucella
Acute vs subacute	Acute = S. aureus, Strep pneumoniae. Subacute = Strep viridans, Enterococci

3 — CLINICAL FEATURES

Symptoms	Fever, chills, night sweats , retrosternal chest pain, dyspnoea/cough, headache/confusion/stroke, N/V/abdominal pain, weight loss, arthralgia/myalgia/back pain
General signs	Fever, peripheral embolism, clubbing , neurological signs. Cardiac murmur 85% (uncommon subacute). Splenomegaly (subacute)
Peripheral stigmata	Splinter haemorrhages (nail fold) · Petechiae (conjunctiva/palate) · Osler's nodes = PAINFUL fingertip nodules · Janeway lesions = PAINLESS palm/sole macules · Roth's spots = retinal haemorrhage + central pallor
DDx	Meningococcal septicaemia, viral illness, hidden abscess, TB, lymphoma, RHD, vasculitis (SLE/RA)

4 — DIAGNOSIS (Modified Duke Criteria) & WORKUP

MAJOR	1. Blood culture +ve : typical organism ×2, OR persistent +ve >12h apart, OR ≥3 +ve over >1h. 2. Endocardial : vegetation on echo OR new valvular regurgitation
MINOR	Predisposing valve/cardiac abnormality · IDU · fever ≥38°C · embolic phenomenon · vasculitic phenomenon · suggestive blood culture · suggestive echo
Diagnosis	2 major, OR 1 major + 3 minor, OR 5 minor
Workup	Blood cultures (key), echo (TTE→TOE) , serial ECG (conduction block = abscess), gated cardiac CT, FDG-PET/CT, brain/spine imaging

5 — ANTIBIOTIC TREATMENT (usually 4–6 wks from 1st negative culture)

Empiric NVE / late PVE (>12m)	Vancomycin + ceftriaxone
Early PVE (<12m)	Vanc + CTX ± gentamicin
Strep	Penicillin / ampicillin / ceftriaxone (± gent if PVE)
Staph	MRSA → vancomycin or daptomycin . MSSA → nafcillin/oxacillin/cefazolin (β-lactam > vanc); rifampin/gent if PVE
Enterococci	Ampicillin + (ceftriaxone or gentamicin) ; VRE → linezolid/daptomycin
Gram-neg / Fungi	HACEK → ceftriaxone/ampicillin/FQ. Pseudomonas → 2 agents. Candida → amphotericin B ± flucytosine/micafungin. Aspergillus → ampho B or voriconazole (+ ophthal consult)
Duration note	IV→PO after ≥10d if appropriate. Uncomplicated right-sided NVE / PCN-sensitive Strep → 2 wks may suffice . Continue anticoagulant unless stroke/ICH/surgery

6 — SURGERY, COMPLICATIONS & PROPHYLAXIS

Surgery indications	Emergent : refractory cardiogenic shock; PVE dysfunction/dehiscence. Urgent : severe HF; penetrating infection (periannular abscess, heart block, fistula); persistent infection (+culture >5d, large/growing vegetation); emboli (recurrent or >10mm + severe AR/MR); S. aureus/fungal/MDR. Cerebral emboli NOT contraindication unless severe stroke/ICH
Complications	Peripheral embolism, stroke , HF, AV block / conduction disorders , acute renal failure
Prophylaxis	Not routine . Need BOTH high-risk condition + high-risk procedure. Conditions : prosthetic valve, previous IE, cyanotic CHD (unrepaired/palliative), 1st 6m after complete repair, transplant valvulopathy. NOT for acquired VHD/BAV/MVP/HCM. Procedure : dental gingival/periapical/mucosal. Regimen : amoxicillin 2g PO 30–60min before (PCN-allergic → cephalexin/azithro/clarithro/doxy)